

THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 2 (2020/2021)

1. (a) Express each of the following numbers in the form $\frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$ and $\frac{a}{b}$ is in its simplest form.

i. $0.666\ldots$

iii. $2.24\overline{13}$

v. $-2.\overline{71}$

ii. $0.4\overline{6}$

iv. $0.818181818\ldots$

vi. $0.0\overline{321}$

- (b) Prove that each of the following numbers are irrational.

i. $\sqrt{3}$

ii. $\sqrt{5}$

iii. $\sqrt{7}$

- (c) Simplify each of the following:

i. $7\sqrt{2} + \sqrt{18}$

iii. $\sqrt{20} + 2\sqrt{45} - \sqrt{80}$

v. $\sqrt{108} - 8\sqrt{3} + 3\sqrt{48}$

ii. $\sqrt{200} + \sqrt{18} - \sqrt{72}$

iv. $3\sqrt{27} - \sqrt{48} + 2\sqrt{75}$

vi. $\frac{3\sqrt{28}}{2\sqrt{175}}$

- (d) Rationalize the denominator in each of the following:

i. $\frac{3}{\sqrt{5}}$

iii. $\frac{3\sqrt{2}}{2+\sqrt{5}}$

v. $\frac{x-y}{\sqrt{x}-\sqrt{y}}$

ii. $\frac{2}{3-\sqrt{3}}$

iv. $\frac{\sqrt{3}-\sqrt{7}}{\sqrt{3}+\sqrt{7}}$

vi. $\frac{h}{\sqrt{x+h}-\sqrt{x}}$

- (e) Rationalize the denominator, and simplify each of the following:

i. $\frac{4}{\sqrt{2}} + \sqrt{8}$

ii. $\frac{8}{3+\sqrt{5}} + \sqrt{45}$

iii. $\frac{\sqrt{3}-\sqrt{7}}{\sqrt{3}+\sqrt{7}} + \sqrt{21}$

2. (a) Simplify each of the following leaving the answer in standard form, $a + bi$, where $a, b \in \mathbb{R}$.

i. $(2 + 3i) - (-7 + 6i)$

iii. $(-2 + 3i)^2$

v. $\frac{2-3i}{1-2i}$

ii. $(3 + 4i) + (6 - 5i)$

iv. $(2 + i)(-1 + 2i)(4 - 3i)$

vi. $\frac{(-2+3i)^2}{1+2i}$

- (b) Given that $z_1 = 3 - 2i$, $z_2 = -1 + i$ and $z_3 = 3 + 2i$, perform each of the following operations, leaving each answer in the form $a + bi$, where $a, b \in \mathbb{R}$.

i. $\frac{2z_1 - 5z_3}{z_3}$

ii. $\frac{\overline{z_2}z_1}{z_3}$

iii. $\frac{(z_2)^2}{\overline{z_1}}$

- (c) Solve each of the following equations.

i. $3 + 5i = (a + bi)(1 + i)$

iii. $(6 + 3i) = 4i - 30(x + yi)$

ii. $(a + i)^3 = 18 + 26i$

iv. $xyi = 3x - y + 12i$

- (d) Given that $\frac{5-2i}{z} = 2 + i$, find z in the form $a + bi$, where $a, b \in \mathbb{R}$.

3. (a) Which sets of ordered pairs are functions in each of the following:

i. $\{(0, -1), (2, 2), (1, -2), (3, 0), (1, 1)\}$

iv. $\{(1, 2), (1, 3), (2, 3)\}$

ii. $\{(1, 4), (3, 4), (7, 3)\}$

v. $\{(1, 2), (2, 3), (3, 4)\}$

iii. $\{(4, 3), (4, 7), (3, 4)\}$

In each of the cases above, illustrate the information on an arrow diagram.

(b) If the function f is defined by

$$f(x) := \begin{cases} 3x + 2 & \text{for } x \geq 0, \\ 5x - 1 & \text{for } x < 0, \end{cases}$$

compute $f(2)$, $f(6)$, $f(-1)$, and $f(-4)$.

(c) If the function f is defined by

$$f(x) := \begin{cases} 2 & \text{for } x < 0, \\ x^2 + 1 & \text{for } 0 \leq x \leq 4, \\ -1 & \text{for } x > 4, \end{cases}$$

compute $f(3)$, $f(6)$, $f(-3)$.

(d) For each of the following functions, find the expression for

$$\frac{f(x+h) - f(x)}{h}$$

i. $f(x) = 4x + 5$

iv. $f(x) = \frac{1}{x}$

ii. $f(x) = x^2 - 3x$

iii. $f(x) = x^2 + 7x - 4$

v. $f(x) = \frac{1}{\sqrt{x}}$

(e) For each of the following functions, find/state the domain

i. $f(x) = 4x + 5$

v. $f(x) = \sqrt{x+4}$

ii. $f(x) = \frac{1}{x}$

vi. $f(x) = \sqrt{3x-1}$

iii. $f(x) = x^2 - 3x$

iv. $f(x) = \frac{1}{\sqrt{x}}$

vii. $f(x) = \sqrt{x^2+4}$

4. (a) Let f and g be two functions. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, specifying the domain in each case:

i. $f(x) = 3x + 4$, $g(x) = x^2 + 1$

iii. $f(x) = \sqrt{x-2}$, $g(x) = 3x - 1$

ii. $f(x) = x^2 - x - 1$, $g(x) = x + 4$

iv. $f(x) = \frac{1}{x-1}$, $g(x) = \frac{2}{x}$

(b) If $f(x) = x^2 - 2$, $g(x) = x + 4$, find $(f \circ g)(4)$ and $(g \circ f)(4)$.

5. (a) Let $g(x) = x^2 - 4$ and $h(x) = \sqrt{x}$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, specifying the domain and range in each case.

(b) For each of the following functions, (i) list its domain and range, (ii) form the inverse function f^{-1} and list the domain and range of f^{-1} .

i. $f = \{(1, 5), (2, 9), (5, 21)\}$

ii. $f = \{(0, 0), (2, 8), (-1, -1), (-2, -8)\}$

(c) Show that the following functions are inverses of each other.

i. $f(x) = 5x - 9$, $g(x) = \frac{x+9}{5}$

ii. $f(x) = \frac{1}{x-1}$ for $x \neq 1$, $g(x) = \frac{x+1}{x}$ for $x \neq 0$.

(d) If $f(x) = 2x + 3$ and $g(x) = 3x - 5$, find

i. $(f \circ g)^{-1}(x)$

iv. $(f \circ g)^{-1}(2)$

ii. $(f^{-1} \circ g^{-1})(x)$

v. $(f^{-1} \circ g^{-1})(4)$

iii. $(g^{-1} \circ f^{-1})(x)$

vi. $(g^{-1} \circ f^{-1})(-10)$

(e) If $f(x) = x - 4$ and $g(x) = \frac{3}{x+1}$, solve the equation

$$(f \circ g)^{-1}(x) = 2.$$